Roll No. | | | | | | | |

Paper Code: TCS 302

END SEMESTER Examination 2024

B.Tech (CSE-Core-All) III Sem

Data Structure with 'C' language.

Time : Three Hours

Maximum Marks :100

NSTRUCTIONS TO STUDENTS

Note:

(i) All questions are compulsory.

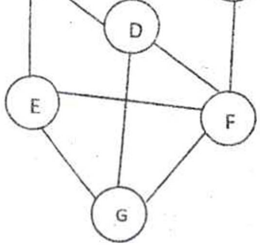
(ii) Answer any two sub questions among a, b & c in each main question.

Q1. (2X10=20 Marks) (CO1, CO3, CO4)

A. Write a c function to delete a node from a circular singly linked list. Consider all the cases.

B. Explain big oh notation. Write an algorithm to find the sum of elements stored in two 2-D arrays, then count total numbers of steps required by the algorithm also predict the nature of the algorithm.

C. Apply Huffman's algorithm to find Huffman's tree and code ,for a=2,b=3,c=4,d=1,e=3,f=4. Also find the minimum weighted path length.



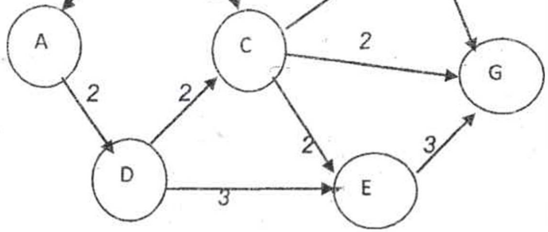
Q3. (2X10=20 Marks) (CO2, CO3, CO5)

A. *Write advantages of an AVL tree. Draw an AVL tree with following keys:*

M,B,C,D,O,N,P,Q,R,E,D

B. *Explain hash collision with an example. Consider a hash table of size (m) 13. Using linear probing technique, insert following keys 112,171,119,122,183,121,192,155,315 , and 229 into the hash table.*

C. Assume that you have a sorted single linked list, with a pointer P pointing the first node of the linked *list. Write a C function to delete all nodes that have duplicate data in that linked list.*

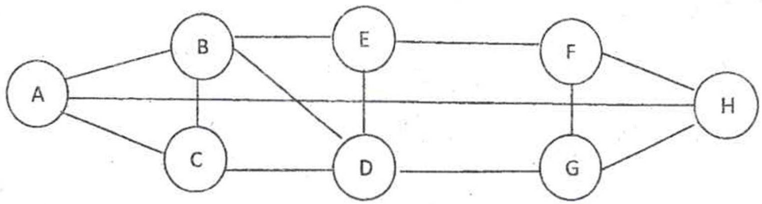


C. *Write an algorithm to delete a node from a binary search tree t. Discuss your algorithm with an example*

Q5. (2X10=20 Marks) (CO3, CO4, CO5)

A. Explain B and B^{+} Tree. Draw a B tree of order 3 using the following keys: 9,11,4,6,13,15,12,14,7,16,17.

B. *What do you mean by a connected graph? Give linked representation and memory representation of following graph.*



C. *Assume that you have a single linked list with pointer P pointing to first node of the linked list. Write a 'C' function split to create two linked lists Q & R. Q contains all elements in odd positions of P and R contains the remaining elements. Your function should not change list pointed by pointer P.*

Roll No. ....

TCS–307

B. TECH. (CSE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

OBJECT ORIENTED PROGRAMMING WITH C++

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions from (a), (b) and (c).

(iii) Each sub-question carries 10 marks.

(iv) Each question carries 20 marks.

(v) For Programming Questions provide typical output.

1. (a) Differentiate Procedural and Object-oriented programming with examples. List the major areas of application of Object-oriented programming. (CO1)

   (b) How namespaces play a vital role in C++ ? Comment in your own words by discussing the advantages of creating namespaces. With the help of a C++ program demonstrate the working of nested namespaces in C++. (CO1)

P. T. O.

(c) Implement a program in C++ to sort the strings in lexicographical order (don't use any predefined string functions) : (CO1)

Typical input : hello

heelo

Typical output : heelo

hello

2. (a) Analyze the given program and determine the output. Provide the output displayed and explain the reason thereof : (CO2)

(i)
```
#include<iostream>
using namespace std;
class P
{
    public:
  P()
  {
  cout <<"\n P's constructor";
  }
  P(P &a)
  {
  cout<<"\n P's copy constructor";
  }
  P& operator= (const P &a)
  {
```

```
        if (this = = &a)
        return *this;

                cout <<"\n P's Assignment Operator";

                return *this;
        }
  };
class Q
{
   P a;
   public:
   Q (P &a): a(a) {cout <<"\n Q's constructor";}
};
   int main ()
   {
        P a;
        Q b (a);
}
```

(ii)

```
#include <iostream>
using namespace std;
class P
{
      public: void foo()
```

```
            {
                cout << "P::f";
            }
            virtual void go()
            {
                cout << "Q::g";
            }
};
class Q: public P
    {
            public: virtual void foo ()
                {
                    cout << "Q::f";
                }
            virtual void go()
                {
                    cout << "Q::m";
                }
};
int main ()
        P x;
        Q y;
        P &ro = y;
        ro. foo();
        ro. go();
        return 0;
}
```

(b) Implement a C++ program to overload binary greater than operator > using friend function only. (CO2)

(c) Create a class named 'BankAccount' with the following data members

Name of depositor (CO2)

Address of depositor

Type of account

Balance in account

Number of transactions

Class 'BankAccount' has a member function for each of the following :

(i) Generate a unique account number for each depositor

For the first depositor, account number will be BA1000, for the second depositor it will be BA1001 and so on.

(ii) Display information and balance of depositor

(iii) Deposit more amount in the balance of any depositor

(iv) Withdraw some amount from the balance deposited.

(v) Change the address of depositor.

After creating the class, do the following operations

(i) Enter the information (name, address, type of account, balance) of the depositors. Number of depositors are to be entered by the user.

(ii) Print the information of any depositor.

(iii) Add some amount to the account of any depositor and then display the final information of that depositor.

3. (a) Create a Base class that consists of private, protected and public data members and member functions. Try using different access modifies for inheriting Base class to the Derived class and create a table that summarizes the above three modes (when derived in public, protected and private modes) and shows the access specifier of the members of base class in the Derived class. (CO3)

   (b) Implement a C++ program to read the content from a file and again write the same content in another file. Assume the name of source file is source.txt and destination file name is dest.txt. (CO3)

   (c) "Multiple inheritance leads to ambiguity." Justify the statement. Also Demonstrate by taking a real time scenario in terms of a C++ code and provide the necessary solution to it. (CO3)

4. (a) Implement a class Publication which has a title. Derive two classes from it. A class Book which has an accession number and a class Magazine which has volume number. With these two as bases, derive the class Journal. Define a function showInfo() in each of these classes. Ensure that the derived class function always invokes the base(s) class function. In main() create object of derived class with an accession number 1000 and a volume number 20. Invoke the showInfo() function for this object. (CO4)

(b) Illustrate Diamond problem in C++ with an example. Discuss the possible solution to resolve it with the help of C++ code.

(c) Differentiate Method Overloading and Method Overriding in C++ using Class by appropriate code. Provide the necessary solution with a proper C++ code that preserves the Property of Method overriding. (CO4)

5. (a) Explain Exception in C++ ? How is it handled in C++ ? Describe all the keywords involved in exception handling by implementing it with a C++ Program. (CO5, CO6)

(b) Construct an Abstract class PathDemo with the following data members and functions : (CO5, CO6)

private :

string str

public :

parameterised constructor to initialize string data member

Pure Virtual functions:

extractFile()

extractDirectory()

Construct a derived class Result that will override all the pure virtual functions such that :

(i) extractFile() – will extract the file name from the given string

Typical Input : C:/geuExam/CS.txt

Typical Output : CS.txt

(ii) extractDirectory() – will extract the directory name from the given path

Typical Input : C:/geuExam/CS.txt

Typical Output : geuExam

(Assume the mode of inheritance as public. Don't assume any methods on your own.)

(c) Discuss the need of STL in C++ with proper examples. Implement a C++ program to swap two numbers using function template.

(CO5, CO6)

Roll No. ....

TCS–308

B. TECH. (CSE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

LOGIC DESIGN & COMPUTER ORGANIZATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Implement full adder using 4 :1 MUX when A and C are selected lines. (CO1)

   (b) Draw and explain 4 bit by 4 bit binary multiplier. (CO1)

   (c) Construct a 4 to 16 decoder using 2 to 4 decoder only. (CO1)

2. (a) Determine the characteristic equation SR flip-flop. Convert SR flip-flop into T flip-flop. (CO2)

   (b) Draw and explain the circuit of universal shift register. (CO2)

   (c) Draw and explain the circuit of MOD 21 asynchronous counter. (CO2)

3. (a) Draw and explain the operation of MCD 5 ring counter. Also determine total number of unused states. (CO3)

(b) Design a synchronous counter (by using T flip-flop) for a count sequence of 0, 2, 5, 7, 0. (CO3)

(c) Design a clocked sequential circuit with two D flip flops, A and B, and one input x. When $x = 0$, the state of the circuit remains the same. When $x = 1$, the circuit passes through the state transitions from 00 to 01 to 11 to 10 and back to 00 and repeats. When it reaches to maximum value output Z becomes 1 otherwise 0. (CO3)

4. (a) State Amdahl's law. If a system has a single bottleneck that occupies 20% of the total execution time, and we add 4 more processors to the system, what would be the overall speedup incorporating the enhancement ? (CO4 & CO5)

(b) Draw the flowchart of booth's algorithm. Perform the following signed number multiplication operation : -7×5. (CO4 & CO5)

(c) Define the following : (CO4 & CO5)

(i) Pipelining

(ii) Von Neumann and Harvard Architecture

5. (a) Define Hit rate and Miss penalty. Suppose that a computer has a processor with two L1 caches, one for instructions and one for data, and an L2 cache. Let 1nsec be the access time for the two L1 caches. The miss penalties are approximately 15 nsec for transferring a block from L2 to L1, and 100 T for transferring a block from the main memory to

L2. For the purpose of this problem, assume that the hit rates are the same for instructions and data and that the hit rates in the L1 and L2 caches are 0.95 and 0.80, respectively. What is the average access time as seen by the processor ? (CO5 & CO6)

(b) Describe memory hierarchy based on parameters like speed, size, complexity, and cost. (CO5 & CO6)

(c) Differentiate between RISC and CISC architecture. Also discuss SIMD array processor. (CO5 & CO6)

Roll No. ....

TMA–316

B. TECH. (CSE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

DISCRETE STRUCTURES AND COMBINATORICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Show that the function :

$$f : \mathrm{R} \to \mathrm{R}$$

defined as $f(x) = 2x + 3$, $\forall x \in \mathrm{R}$ is both one-one and onto function also find its inverse. Also find function $g : \mathrm{I} \to 1$ defined as $g(x) = x^2 + 1, \forall x \in \mathrm{I}$ is a bijection map or not. (CO1)

(b) Let R be a binary relation defined as :

$$\mathrm{R} = \{(a, b) \in \mathrm{R}^2 : |a - b| \leq 3\}$$

Determine whether R is reflexive, symmetric, antisymmetric and transitive. (CO1)

P. T. O.

(c) Draw the Hasse diagram of D_{30} (positive divisors of 30) and also find its complement. Is it a complemented lattice ? (CO1)

2. (a) A coffee a sampler claims that he can distinguish between a cup of instant coffee and a cup of percolator coffee 75% of time. It is agreed that his claim will be accepted if he correctly identifies at least 5 of the 6 cups. Find his chance of having the claim : (CO2)

(i) accepted

(ii) rejected, when he does have the ability he claims.

(b) The random variable Y has probability density function $f(y)$ given by : (CO2)

$$f(y)=\begin{cases} ky(a-y), & 0\leq y\leq 3 \\ 0 & , \text{otherwise} \end{cases}$$

where k and a are positive constants.

(I) (i) Explain why $a \geq 3$

(ii) Show that $k=\dfrac{2}{9(a-2)}$

Given that $E(Y)=1.75$

(II) Show that $a=4$ and write down the value of k.

(c) If X is a Poisson variate such that : (CO2)

$$P(X=2)=9P(X=4)+90P(X=6)$$

find λ (the mean for X).

3. (a) Prove the distributive law :

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

using truth table. Also prove the associative law $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$. (CO3)

(b) Prove by the principle of mathematical induction : (CO3)

$$n! > 2^n \text{ for all } n > 4$$

(c) Check the validity of the following argument : (CO3)

"If it is Saturday today, then we play soccer or basketball. If the soccer field is occupied, we don't play soccer. It is Saturday today, and the soccer field is occupied. Therefore, we play basketball or volleyball."

4. (a) Show that the set :

$$G = \{1, 2, 3, 4, 5, 6\}$$

forms an Abelian group with respect to multiplication modulo 7. Is it a cyclic group. (CO4)

(b) Define Permutation. How many numbers greater than a million can be formed with digits 2, 3, 0, 3, 4, 2, 3 ? (CO4)

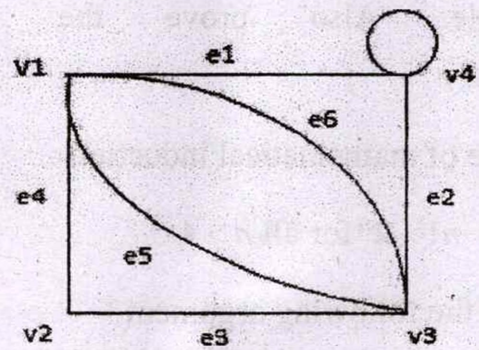
(c) Solve the recurrence relation : (CO4)

$$a_n - 7a_{n-1} + 12a_{n-2} = n.4^n$$

5. (a) (i) Draw the undirected graph G corresponding to adjacency : (CO5)

$$\begin{bmatrix} 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & 0 \\ 3 & 1 & 0 & 0 \\ 2 & 0 & 0 & 1 \end{bmatrix}$$

(ii) Find the adjacency matrix and the incident matrix of the multigraph shown in Fig.



(b) Define with examples : (CO5)

(i) Regular Graph

(ii) Binary Tree

(iii) Path

(iv) Complete Graph

(v) Bipartite Graph

(c) Solve by the method of generating functions the recurrence relation :

$$a_r - 2a_{r-1} + a_{r-2} = \frac{1}{4}2r$$

$r \geq 2$ with the boundary condition $a_0 = 2$ and $a_1 = 1$. (CO5)

Roll No. ....

TCS–331

B. TECH. (CSE)/BIOTECH (THIRD SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023

FUNDAMENTAL OF IOT

Time : Three Hours

Maximum Marks : 100

Note.: (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the data overload and analysis challenge in IoT system for potential smart city traffic management system. Also discuss the impact and potential mitigation approach. (CO1 & CO2)

(b) Compare key components used in automation and Internet of Things. (CO1 & CO2)

(c) Explain the following terminologies : (CO1 & CO2)

(i) Decryption

(ii) Firmware

P. T. O.

2. (a) Consider yourself an IoT architect, you have to design a schematic layout with respect to various layers in IoT system for a potential gesture control oar application. (CO2 & CO3)

(b) What are the various nodal capabilities (any *five*) of an IoT system ? Explain with suitable brief examples. (CO2 & CO3)

(c) Explain the following terminologies : (CO2 & CO3)

(i) Actuator

(ii) Bluetooth

3. (a) Explain the working of RFID module using suitable diagram. Also explain its different types. (CO3 & CO5)

(b) Compare the working of QR codes and Bar codes. Also remark on their suitability criteria for selection in particular application (CO3 & CO5)

(c) Explain the following terms in brief : (CO3 & CO5)

(i) MAC Address

(ii) IP Address

4. (a) Describe Web of Things (WoT) . Considering yourself an IoT architect suggest key components and working for a design to implement WoT for smart farming. (CO4)

(b) Explain key components of wireless sensor network. Also explain sensing node, connecting node and networking node. (CO4)

(c) Explain various clustering methods with respect to suitable example of some real time applications. (CO4)

5. (a) Explain various identity management models of Internet of Things using suitable examples. (CO6)

(b) Compare user centric identity systems and device centric identity systems. (CO6)

(c) Design a model application for smart card-based student entry and attendance system for our university. Define the resources, features, and block diagram of the proposed system. (CO6)

Roll No. ....

TCS–332

B. TECH. (CSE & SPL) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

FUNDAMENTAL OF INFORMATION SECURITY AND BLOCKCHAIN

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is the use of penetration testing ? Demonstrate the working procedure of penetration testing. (CO1)

(b) What is a blockchain ? Describe different types of blockchains along with their uses. (CO1)

(c) Describe the following terminologies : (CO1)

Cyber vandalism and hacktivism.

2. (a) What is the use of an access control mechanism ? Design an access control scheme for a smart hospital management system. (CO6)

(b) What is the use of a private blockchain ? Examine the security of a private blockchain. (CO6)

(c) What is the adverse impact of cross-site scripting (XSS) attacks ? What are the differences between reflected XSS and stored XSS attacks ? (CO6)

3. (a) How would you compare 2-factor user authentication and 3-factor ul authentication schemes ? (CO3)

(b) In the blockchain, what are the different fields of a block ? Draw the design of a block. Which field is used for which purpose ? Give an estimate of these fields' size in bits. (CO3)

(c) How would you compare the following consensus algorithms ? (CO3)

Proof-of-Capacity and Proof-of-Weight.

4. (a) What is Paxos ? What are the different defined roles in Paxos ? What are the uses of these roles ? (CO5)

(b) Describe the following properties of a blockchain : (CO5)

Unanimous, anonymous, secure, distributed and timestamped

(c) Write down a solidity program to create a simple storage with a clear explanation. (CO5)

5. (a) What is smart contract ? How does it work ? How does it differ from a normal contract ? (CO2)

(b) What is the use of netcat ? Design a chat application using the netcat tool. (CO2)

(c) How would you compare the following types of addresses in TCP/IP ? IP address and port address. (CO2)

Roll No. ....

TCS–341

B. TECH. (CSE) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023

PYTHON PROGRAMMING FOR COMPUTING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the following concepts of python with example : (CO1 & CO2)

   (i) range() function

   (ii) break and continue statements

   (iii) pass statements

   (b) Design a menu driven Program in Python to perfom the following tasks : (CO1 & CO2)

   (i) To check if an entered year is leap year or not ?

   (ii) To calculate a GCD of two numbers

   Support the program with appropriate functions for each of the above tasks.

   (c) Explain any ten string methods with example. (CO1 & CO2)

2. (a) Discuss the python data structure set methods with code snippet. (CO2 & CO3)

(b) Design a menu driven Program in Python for the following operations on Queue (List Implementation of Queue with maximum size MAX) : (CO2 & CO3)

(i) Insert an Element on to Queue

(ii) Delete an Element from Queue

(iii) Demonstrate Overflow and Underflow situations on Queue

(iv) Display the status of Queue

(v) Exit

Support the program with appropriate functions for each of the above operations.

(c) For a given list num [45, 21, 14, 65, 97, 75]. Develop a python program to perform following operations and write the output :

(i) To replace all the numbers divisible by 3 with "TTT"

(ii) To replace all the numbers divisible by 5 with "FFF"

(iii) To replace all the numbers divisible by both 3 and 5 with "TTTFFFF"

3. (a) Discuss any *five* types of exception handling in python with example. (CO3)

(b) Runs scored by Virat Kohli is last 8 matches is as follows : 101, 88, 0, 95, 103, 16, 55, 85. These runs are stored in a file Virat.txt. Develop a python program to perform following operations on file Virat.txt. (CO3)

(i) Read a data from the file and find the highest run scored by Virat Kohli

(ii) Read a data from the file, arrange the runs scored by Virat Kohli in descending order and write the result into desc.txt

(iii) Read a data from the file and calculate the total run scored by Virat Kohli

(c) "A Christmas celebration is arranged in a school from 24-12-2023 to 25- 12-2023. Various events are organized in a school; members interested in attending sports and cultural events have to send an email to sports@gamil.com and cultural@gmail.com, respectively. The last date for registration is 20-12-2023. For any queries, feel free to contact +91- 9999888822 or send an email to queries@gmail.com". The sentence is stored in a variable named str. Develop a Python program to perform the following operations and write the output after each operation : (CO3)

(i) To extract date from the str

(ii) To extract email from the str

(iii) To extract mobile number from the str

(iv) To find all six-character words in a string

(v) To find all words that contain letters both a and i

4. (a) Consider the dictionary given below : (CO3)

exam_data = {

'name' : ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score' : [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts' : [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify' : ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']

} (CO4 & CO5)

(i) Develop a Pandas program to create and display a DataFrame from a specified dictionary data

(ii) Develop a pandas program to fill missing value of score column with mean value of the score

(iii) Develop a Pandas program to select the rows where the number of attempts in the examination is greater than 2

(iv) Develop a Pandas program to calculate the sum of the examination attempts by the students

(v) Develop a Pandas program to replace the 'qualify' column contains the values 'yes' and 'no' with True and False

(b) The ICC World Cup match between India and South Africa was played on 5th November, 2023. India won by 243 runs. The wickets taken by Indian bowlers are as follows :

Mohammed Siraj : 1, Ravindra Jadeja : 5, Mohammed Shami : 2, Kuldeep Yadav : 2.

Plot the pie chart for wickets taken using the Matplotlib library with the following parameters : (CO4 & CO5)

(i) Add labels to the pie chart with respective bowlers names.

(ii) Start the first wedge at 90 degrees.

(iii) Pull the highest wicket wedge 0.1 from the center of the pie.

(iv) Specify a new color for each wedge.

(v) Add a legend and shadow.

(c) Illustrate the following concepts with python program : (CO4 & CO5)

(i) List Comprehensions

(ii) File handling write mode vs append mode

(iii) Regular expression search() method vs findall() method

5. (a) Develop a Python program to perform following task using Lambda :

(CO5 & CO6)

(i) To read 10 integer numbers into list and filter even numbers using Lambda

(ii) To read 5 numbers into a list and cube every number in a given list of integers using Lambda

(iii) To check whether a given string is a number or not using Lambda

(b) Create a Python class called Laptop with Name, Model Number, Processor and Price as variables within it. Develop a Python program to create n Laptop objects and print the Name, Model Number, Processor and Price of these objects with suitable headings. (CO5 & CO6)

(c) Illustrate the following concepts with python program : (CO5 & CO6)

(i) Overloading and Overriding

(ii) Multiple Inheritance

Roll No. ....

TCS–342

B. TECH. (AI AND DS) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023

INTRODUCTION TO STATISTICAL DATA SCIENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the five V's in big-data, explain significance of various 'V'. (CO1)

   (b) Define : (CO2)

   (i) Layered approach to Data Science

   (ii) Pyramid of Data Science

   (c) Discuss the W. L. L. N. with respect to obtaining Empirical Probability, how is this model influenced by n = number of total trials, k = number of classes or partitions of sample space. (CO3)

P. T. O.

2. (a) State and prove Central Limit Theorem, why does it denote weak convergence. (CO3)

   (b) Define random variable and its need in data modelling. Starting with random variable formulate : (CO1)

   C. D. F. P. D. F. and P. M. F.

   (c) Formulate some mode to test when collected data behaves one of the following random variables : Uniform, Exponential, Geometric, Binomial. (CO2)

3. (a) Give generalized statistical estimate of mean-function and auto-correlation function for real-time continuous data. Now, define time-series and depict these statistical estimates (mean and autocorrelation function) for time-series. (CO1)

   (b) Draw and give statement of Bayes' theorem, explain various attributes of Bayes' theorem useful in statistical decision models. (CO3)

   (c) What part of time-series data is Outlier ? Describe intra-class and cross-class outliers. Give at least 2 methods for outlier detection in a time-series data. (CO2)

4. (a) Determine if $g(x)$, as given below, are valid Distribution Functions (D. F.) : (CO3)

$$g_1(x) = 0.5\,\text{exp.}\,(-0.5(x)), 0 < x < \infty;$$

$$g_2(x) = \frac{1}{2N}, 0 \le x \le N/2$$

   Enumerate the main properties of Cumulative Distribution Function – $F_x(x)$. (CO3)

(b) An urn has 2000 apples, 100 pencils and 3000 watches. Given 200 items are drawn at once from the urn, determine probability of drawing 160 apples, 20 pencils and 20 watches. Define when can we apply Negative binomial distribution as data model. (CO3)

(c) Illustrate/plot Gaussian random variable by stating its p.d.f. Also, enumerate at least 6 properties that distinctly map for Gaussian random variable. (CO1)

5. (a) Given 3 coins that are tossed separately denote random variable X, Y and Z; if O = X + Y + Z denotes random variable, then determine the mean, variance and distribution function of Z. (CO2)

(b) Discuss binary hypothesis testing and the errors resulting. Discuss use cases of this model in : (CO3)

(i) Spam filter for e-mail

(ii) Detecting tumor from imaging data

(c) Discuss : (CO3)

(i) Pearson correlation for time series data,

(ii) Spearman test for correlation.

How does one of these influence Linear regression as generative data model ?

Roll No. ....

TCS–343

B. TECH. (CSE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

MATHEMATICAL FOUNDATIONS FOR ARTIFICIAL INTELLIGENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Find a linear map :

$$F : R^3 \rightarrow R^4$$

whose image is spanned by (1, 2, 0, – 4) and (2, 0, – 1, – 3). [Here, R is the real field]. (CO1)

(b) Test for consistency and if possible, solve the following systems of equations by rank method : (CO1)

$$x - y + 2z = 2$$

$$2x + y + 4z = 7$$

$$4x - y + z = 4.$$

(c) Let : (CO1)

$$F : R^4 \rightarrow R^3$$

be defined by

$$F(x, y, z, t) = (2x + 3y - z + 2t,\ x - 5y + 6t, 2y + z + t).$$

Find the matrix A which represents F using the usual bases E_4 and E_3.

[Here, R is the real field.]

2. (a) Consider the following basis of Euclidean space :

$$R^3 : \{v_1 = (1, 1, 1), v_2 = (0, 1, 1), v_3 = (0, 0, 1)\}.$$

Use the Gram-Schmidt algorithm to transform $\{v_i\}$ into an orthonormal basis $\{u_i\}$ of R^2. (CO2)

(b) Consider $u = (0, 1, -2, 5)$ in R^4. Find a basis for the orthogonal complement of u. (CO2)

(c) Find cos θ for the angle θ between : (CO2)

$$f(t) = 2t - 1$$

and $$g(t) = t^2$$

in the vector space V of polynomials with inner product

$$\langle f, g \rangle = \int_0^1 (t) g(t)\, dt$$

3. (a) Perform the Cholesky decomposition for the following matrix : (CO3)

$$\begin{bmatrix} 6 & 15 & 55 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix}$$

(b) Find the singular value decomposition of a matrix : (CO3)

$$A = \begin{bmatrix} -4 & -7 \\ 1 & 4 \end{bmatrix}$$

(c) Find the eigen decomposition and diagonalize of the given matrix : (CO3)

$$\begin{bmatrix} 3 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix}$$

4. (a) Calculate the second-degree Taylor polynomial of : (CO4)

$$f(x, y) = e^{-(x^2+y^2)}$$

at the point (1, 2).

(b) Let : (CO4)

$$f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$$

which is define anywhere except at the origin. Find the gradient

$$F = \nabla f .$$

(c) Find the linearization $L(x, y)$ of $y\sqrt{x}$ at the point (9, – 2). (CO4)

5. (a) Let X be a continuous random variate with probability density function : (CO5)

$$f(x) = \begin{cases} ax, & 0 \le x \le 1 \\ a, & 1 < x \le 2 \\ 3a - ax, & 2 < x \le 3 \\ 0, & \text{elsewhere} \end{cases}$$

(i) Determine the constant a.

(ii) Compute $P(X \le 1.5)$.

(b) The mean yield for a one-acre plot is 662 kilos with a standard deviation of 32 kilos. Assuming a normal distribution, how many one-acre plots in a batch of 1000 plots would you expect to have yield (i) over 700 kilos, (ii) below 650 kilos, and (iii) what is the lowest yield of the best 100 plots ? (CO6)

(c) In answering a question on a multiple choice test a student either knows the answer or he guesses. Let p be the probability that he knows the answer and $1 - p$ the probability that he guesses. Assume that a student who guesses the answer will be correct with probability 1/5, where 5 is the number of multiple-choice alternatives. What is the conditional probability that a student knew the answer to a question given that he answered it correctly ? (CO6)

Roll No. ....

TCS–351

B. TECH. (CSE) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023

FUNDAMENTALS OF CLOUD COMPUTING AND BIG DATA

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) VMware Carbon Black, an arm of VMware, used managed service Amazon Kinesis Data Streams to build a scalable, cost-effective data streaming solution that enhances operations and supports growth. Evaluate Cloud Computing On Demand Services from perspective of above mentioned organization. (CO1)

(b) Database company ClickHouse Inc. developed and launched ClickHouse Cloud in 1 year using AWS services, including Amazon EKS, Amazon EC2 Spot Instances, and Savings Plans. Evaluate privacy of Cloud Computing from perspective of above mentioned organization. (CO1)

P. T. O.

(c) Biz2Credit accelerated the scaling of Kubernetes clusters, automated processes, optimized server instances, and improved uptimes by using Amazon EKS, resulting in improved uptimes, lower costs, and faster customer onboarding. Evaluate Cloud Computing Data Storage Services from perspective of above mentioned organization. (CO1)

2. (a) WirelessCar virtually removed cold starts and improved the performance of its connected mobility services at no extra cost by embracing AWS Lambda SnapStart for Java. Analyze SAP Cloud Platform from perspective of above mentioned organization. (CO2)

(b) Katalon used Amazon MSK and AWS Lambda to build a managed, serverless infrastructure that sped up time to market and cut costs for customers. Analyze the Backend Development Roadmap from the perspective of above mentioned organization. (CO2)

(c) WorkApps transformed its broadcast messaging solution, follow.us, using Amazon SQS, AWS Media Services, and Amazon Translate, ensuring seamless integration, high scalability, and the ability to broadcast millions of messages in second. Analyze the Frontend Development Roadmap from the perspective of above mentioned organization. (CO2)

3. (a) Sony India Software Centre scaled its Cloud Data Platform by migrating to Amazon Redshift RA3 instances, boosting query performance, reducing management time, and increasing employee satisfaction. Analyze Eucalyptus Architecture, Features, Operation Modes and Advantages from the perspective of above mentioned organization.

(CO3)

(b) Zoomcar supported its growing data volumes by implementing data analytics services from AWS, including Amazon Redshift, Amazon EMR, Amazon S3, Amazon Athena, and AWS Glue, empowering faster decision-making, efficient operations, and personalized customer experiences. Analyze hypervisor from the perspective of above mentioned organization. (CO3)

(c) YOUGotaGift consolidated its infrastructure to improve the reliability and performance of its platform and improve innovation. This has resulted in new products, partnerships, and exponential growth. Analyze Amazon Machine Image Types and Properties from the perspective of above mentioned organization. (CO3)

4. (a) Using fully managed AWS services, financial technology company ekonoo SA achieved regulatory approval for its pension management solution and can focus on delivering value to its customers. Analyze Big Data and data Mining Applications, Advantages and Disadvantages from the perspective of above mentioned organization. (CO4)

(b) Luxury fashion company Tapestry built a scalable IaC platform to seamlessly deploy modernized workloads in a nimble, consistent, and repeatable manner for security and governance, helping the company reduce deployment times, maintain security and compliance protocols, and drive innovation. Analyze the types of Machine Learning Algorithms, Life Cycle and Advantages from the perspective of above mentioned organization. (CO4)

(c) Fintech lending platform Aro increased its efficiency using managed search clusters on Amazon OpenSearch Service, creating more capacity to innovate. Analyze the Federated Learning and approaches to sentiment analysis, Types and Challenges from the perspective of above mentioned organization. (CO4)

5. (a) Travis Perkins built on Amazon Web Services (AWS) as part of its strategic effort to create an agile and innovative business and to free data siloed in legacy infrastructure for better insights. Analyze the Data Fusion, Data Integration Techniques and Natural Language Processing (NLP) Components, Phases and Applications from the perspective of above mentioned organization. (CO5, CO6)

   (b) Moody's Senior Vice President of Cloud Engineering, Divya Elaty, discusses how Moody's leverage AWS services like AWS Glue, Amazon Dynamo DB, Amazon S3, and AWS Step Functions to build modern data solutions and custom in-house tools to meet the company's data needs. Analyze AWS Lambda, AWS Elastic Beanstalk, Amazon Route 53, AWS VPC, and Amazon Glacier Services from the perspective of above mentioned organization. (CO5, CO6)

   (c) FOX has improved its marketing and advertising insights and ML capabilities by building a unified data solution using Amazon S3, Amazon Rekognition, and other cloud technology. Analyze AWS Elastic Disaster Recovery and SWOT analysis from the perspective of above mentioned organization. (CO5, CO6)

Roll No. ....

TCS–392

B. TECH. (CSE AND ALL SPECIALIZATIONS)
(THIRD SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023

INTRODUCTION TO CRYPTOGRAPHY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Provide a neat diagram classifying the attacks under each of Passive and Active category. Explain each attack with a typical example. (CO1)

(b) Identify the security services required to prevent attacks and explain each. (CO1)

(c) (i) List out the rules for substituting a pair of plaintext letters (converting plaintext to ciphertext) using a Key matrix in Playfair cipher. (CO1)

P. T. O.

(ii) Construct a Playfair matrix with the key "largest". Further, using the key matrix constructed, convert the message "Must see you over Cadogan West tonight" to ciphertext. (CO1)

2. (a) Explain the features of S-DES. Further, explain with the help of generic block diagram Encryption, Decryption and Key Generation stages of S-DES. (CO2)

(b) (i) Provide a block diagram depicting generation of sub-keys K1 and K2 in S-DES using the Key K.

(ii) Consider a key K = 1110001011, P10 = {3, 5, 2, 7, 4, 10, 1, 9, 8, 6} and P8 = {6, 3, 7, 4, 8, 5, 10, 9}. Generate the sub-keys K1 and K2 of S-DES, using the steps shown in block diagram above. Clearly show output of each step and explain. (CO2)

(c) (i) Provide the block diagram and explain the concept of Double DES. Make use of equations to show encryption and decryption.

(ii) Why do you think the following equation is not true in case of Double DES with keys k1 and k2 ? Justify. (CO2)

$$Ek2[Ek1(P)] = Ek3[P]$$

3. (a) Sender A would like to communicate with recipient B. With the help of neat figure, explain the dialogues and messages exchanged between KDC, sender and recipient to understand symmetric key to be used. (CO3)

(b) (i) What are the usage of Pseudo Random Numbers ? Further, Identify the desired properties of Pseudo Random Numbers.

(ii) Explain the Linear congruential method of generation of Pseudo Random Numbers with the selection criteria of parameters. (CO3)

(c) (i) Provide the steps/pseudo code involved in calculating GCD using Euclidean Algorithm.

(ii) Calculate the GCD of the following numbers using Euclidean Algorithm. Provide the table with variables A, B and R and show the values at each step.

- gcd (4655, 12075)
- gcd (24140, 16762) (CO3)

4. (a) Consider a sender A and recipient B communicating with each other. Each of them possess a Public Key and a Private Key. With the help of nest figures for each case, explain how to achieve the desired property using Public and Private keys.

- Sender A sending message to B wants to achieve only confidentiality.
- Sender A sending message to B wants to place authentication in the message. The message is not confidential.
- Sender A sending message to B wants to achieve confidentiality as well as place authentication in the message. (CO4)

(b) (i) Provide the steps involved in Key generation, Encryption and Decryption using RSA Algorithm.

(ii) Encrypt and Decrypt the message M using RSA given following details :

- $p = 5; q = 11, e = 3; M = 9$
- $p = 11; q = 13, e = 11; M = 7$ (CO4)

(c) Message Authentication Code (MAC) can be appended either after encryption or prior to encryption. Propose with neat figures and explanation the following, using two shared keys k1 and k2.

- Message authentication and confidentiality where authentication is tied to plaintext.
- Message authentication and confidentiality where authentication is tied to ciphertext. (CO4)

5. (a) Elaborate the concept of Distributed Intrusion Detection with the nest figure depicting an architecture and role of components involved.

(CO5, CO6)

(b) What do you understand by the term Password management system ? Elaborate the approaches that can be adopted for selecting the passwords. Identify the advantages and limitations of each of the approaches. (CO5, CO6)

(c) (i) Identify the computer crime based on the role that the computer plays in the criminal activity along with Challenges faced by Law Enforcement Agencies.

(ii) Classify the main types of IPs where legal protection is available and briefly explain each. (CO5, CO6)

120